

Kartik Joshi

linkedin.com/in/kartikjoshi23 | github.com/KartikJoshi23 | kartik-joshi.vercel.app
+971-565582009 | kartujoshi2002@gmail.com | Dubai, UAE

Summary

AI Engineer spanning graph deep learning, reinforcement learning, real-time systems, and full-stack deployment – most recently building an end-to-end graph neural network platform for financial and procurement crime detection. Smart India Hackathon Finalist, ICC NIUM Global Top 10, and IJANA-published ML researcher. Completing a Masters in AI with Business at SP Jain Dubai, targeting applied AI engineering roles in the UAE market.

Education

SP Jain School of Global Management

Masters in AI with Business – 2-Year Postgraduate Programme, CPI: 3.6/4.0

Dubai, UAE

2025 – 2027

Parul University

B.Tech in Computer Science & Engineering (AI), CGPA: 8.04/10

Vadodara, India

2020 – 2024

- Selected as University Representative for the National Entrepreneurship Challenge at IIT Bombay.

Experience

Lead AI Engineer

VaultxLedger Technology Solutions

Apr 2024 – Jun 2026

Vadodara, India

- Designed and demonstrated a real-time computer vision pipeline for Vadodara Rural Police – a working ANPR prototype built in PyTorch and OpenCV achieving sub-100ms inference latency; received a Certificate of Appreciation from IPS Rohan Anand for the prototype demonstration.
- Directed a team of 6 engineers through computer vision pipeline architecture, hardware integration, and direct coordination with IPS-ranked law enforcement officers – delivering the prototype on schedule without external QA support.
- Converted the engagement into VaultxLedger's first government client relationship, translating a hackathon prototype into a confirmed pipeline through the PIERC incubation program.

Incubation Program Associate

PIERC Startup Incubator

Dec 2023 – Mar 2024

Vadodara, India

- Developed go-to-market strategy and investor pitch for VaultxLedger during PIERC Cohort-2; secured incubation support and progressed to a confirmed government client engagement with Vadodara Rural Police.

Projects

CollusionGraph – Graph Deep Learning for Financial & Procurement Collusion Detection

2026

PyTorch, PyTorch Geometric (GATv2, GraphSAGE, R-GCN), PyGOD, FastAPI, React, TypeScript, AWS, NVIDIA NIM

- Built an end-to-end graph deep learning system unifying six public datasets across financial (money laundering) and procurement (bid-rigging) domains into one domain-agnostic graph schema, screening for coordinated networks that no single-record model can detect.
- Combined supervised GNNs (GATv2, GraphSAGE, R-GCN), unsupervised graph anomaly detectors (DOMINANT, GAE), and classical baselines into an isotonic-calibrated ensemble – recovering 92% of planted bid-rigging cartels (vs. 12% for a non-learning baseline) on a 488K-node, 1.45M-edge unlabeled network where classical models could not run at all.
- Shipped as a full product: an explainability layer (PGExplainer + 9 FATF/OECD motif patterns), a tool-grounded GenAI Investigator Copilot passing a 24-question golden-test gate, a React/TypeScript console, and a FastAPI read-only serving layer enforcing a strict train/serve trust boundary.

AlgoViz – Real-Time Algorithmic Trading Dashboard

2025

Python, Streamlit, Plotly, NumPy, SciPy, scikit-learn, Docker, Binance WebSocket API

- Engineered a live trading dashboard streaming BTC/USDT data via the Binance WebSocket API at 50+ trades/sec, computing 9 market microstructure features (VWAP, TWAP, order book imbalance, price velocity, rolling volatility) for signal generation.
- Designed a custom LSTM-Attention network implemented purely in NumPy – zero external deep learning framework dependencies – achieving sub-100ms inference on streaming tick data.
- Delivered a 7-component ensemble predictor with a full P&L backtesting engine, Dockerized deployment, and 45 pytest unit tests.

Smart City Traffic Management – Multi-Agent Reinforcement Learning

2026

Python, React 18, Vite, NumPy, Pandas, TOPSIS

- Designed a Multi-Agent Reinforcement Learning (MARL) system for adaptive traffic signal control across a 16-intersection Dubai grid, training decentralized Q-Learning and SARSA agents over 60,000 steps with a 6-factor reward function balancing queue reduction, throughput, and emissions.
- Built a custom simulation environment with a 3-protocol coordination layer (Green Wave, Conflict Resolution, Corridor Balancing) and a Chaos Mode engine stress-testing sandstorm conditions, corrupted sensors, and emergency vehicle preemption.
- Benchmarked agent policies via WSM and TOPSIS multi-criteria frameworks across 4 weighted metrics; delivered results through a React 18 dashboard with live grid-state animations and Q-value Explainable AI (XAI) panels.

Smart Contract Security Scanner

2026

Python, FastAPI, React 18, TypeScript, Gemini 2.5 Flash, Docker

- Built a full-stack smart contract security scanner for Dubai's DIFC Web3 ecosystem, integrating a React 18 frontend, a FastAPI REST backend, and a Google Gemini 2.5 Flash LLM inference pipeline.
- Implemented a 0–100 risk scoring engine with line-level Solidity vulnerability mapping across 10+ categories (reentrancy, overflow, access control) and automated fix generation across 6 blockchain networks via the Alchemy API.

Publications

AI Cloud Service Recommendation System – IJANA Journal, ASICS Conference

2024

- Co-designed the ML pipeline and model evaluation framework for a cloud service recommendation system, achieving 70%+ classification accuracy across 6 user-defined parameters on a proprietary dataset of 1,000+ records.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS
- **AI & Machine Learning:** PyTorch, TensorFlow, scikit-learn, OpenCV, MediaPipe, NumPy, Pandas, SciPy, Reinforcement Learning, LSTM & Attention, Deep SVDD, Explainable AI (XAI)
- **LLM & Generative AI:** Google Gemini API, LLM Inference Pipelines, Prompt Engineering, Responsible AI, AI Chatbot Development
- **Web & DevOps:** React, Vite, FastAPI, SQLAlchemy, Docker, Git, GitHub Actions, REST APIs, Jupyter Notebooks
- **Developer Tools:** Cursor, GitHub Copilot, Claude (Anthropic), Git/GitHub
- **Domains:** Real-Time Systems, Edge Deployment, MLOps, Graph Machine Learning, Web3 & Blockchain, Smart City Systems, Data Analytics

Achievements

- Global Top 10 – ICC NIUM Hackathon, Web3 & Blockchain (International)
- Finalist – Smart India Hackathon 2023 (National, 1M+ participants)
- Winner – Let's Hack India Tour, Parul University

Leadership & Activities

- **AI Presenter & Product Demonstrator**, Hackathons, Incubators & Law Enforcement Engagements, India (2023 – Present): Delivered live technical demonstrations of AI systems to IPS-ranked law enforcement officers, incubator panels, and international hackathon judges across PIERC, IIT Bombay NEC, Smart India Hackathon, and ICC NIUM.
- **Marketing Lead & Event Head**, CDC FunFest, Parul University, Vadodara, India (2022 – 2023): Managed end-to-end operations and led external communications for a cross-departmental event team.
- **Bal Sanchalak (Youth Leader)**, SMVS Swaminarayan Sanstha, Gujarat, India (2019 – Present): Oversaw mentorship programmes for youth cohorts, developing leadership and facilitation skills.

Languages

English (Professional) • Hindi (Conversational) • Gujarati (Native)